VII Tutorial

Stationarity, model identification and unit-root testing

Exercise 1. Stationarity of ARMA models

Let ε_t be i.i.d., with mean zero and variance $\sigma^2 > 0$. Assume that ε_t is independent of past values of y. Consider the following models:

(a) $y_t - 0.5y_{t-1} = \varepsilon_t$;

(b) $y_t + 0.9y_{t-1} = \varepsilon_t$;

(c) $y_t + 1.9y_{t-1} + 0.88y_{t-2} = \varepsilon_t + 0.5\varepsilon_{t-1}$.

For each model, write the autoregressive polynomial $\phi(z)$, calculate its roots and determine whether a weakly stationary solution exists under the innovation assumption above. State the root condition you use. For model (c), explain the role of the moving-average term and distinguish the stationarity condition from invertibility.

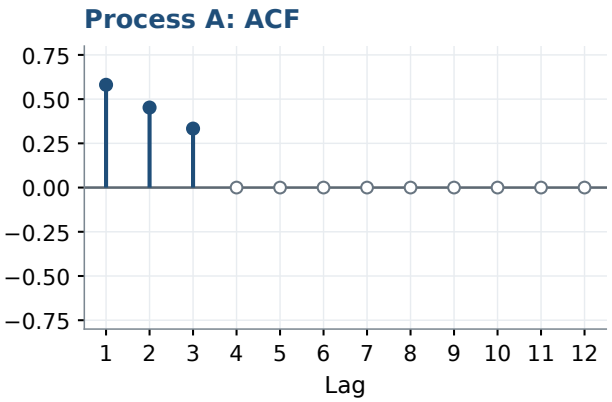
Exercise 2. Recognising AR and MA processes

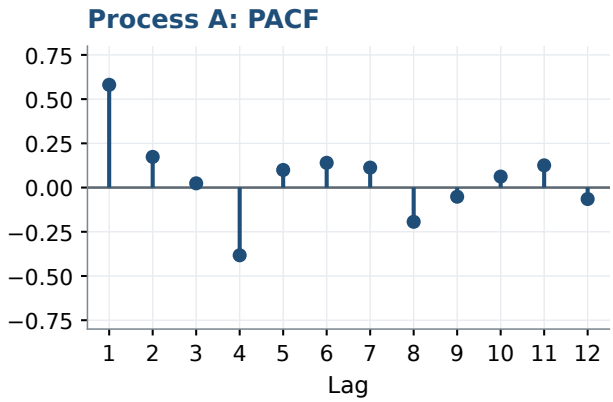
The plots on this and the following page show the **theoretical population ACF and PACF** for four processes. Each is either a causal stationary AR(p) or an invertible MA(q), with a nonzero coefficient on its highest-order lag. Lag zero is omitted. **Filled dots indicate nonzero correlations; hollow circles indicate exact zeros.** All panels use the same vertical scale. There are no sample confidence bands.

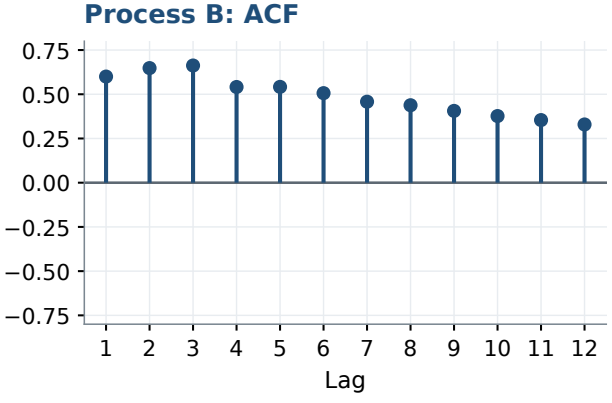
(a) Classify each process as AR or MA and identify its order. Justify each answer using the cutoff in one function and the tail in the other.

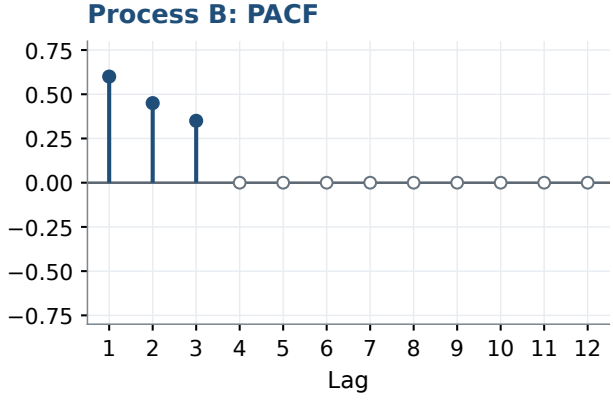
(b) Can the ACF alone generally reveal the order of an AR process? Explain what additional information the PACF provides.

(c) How would your interpretation change if these were sample ACFs and PACFs estimated from a finite dataset?

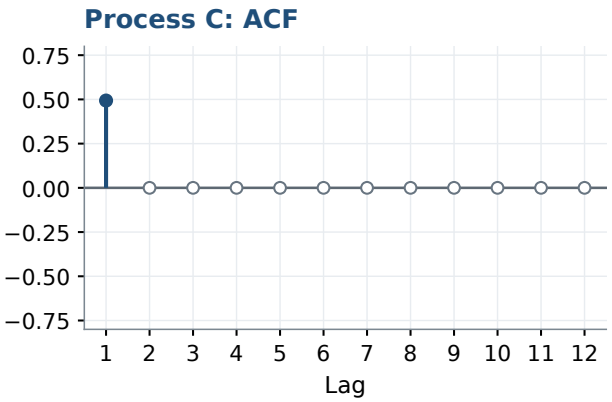


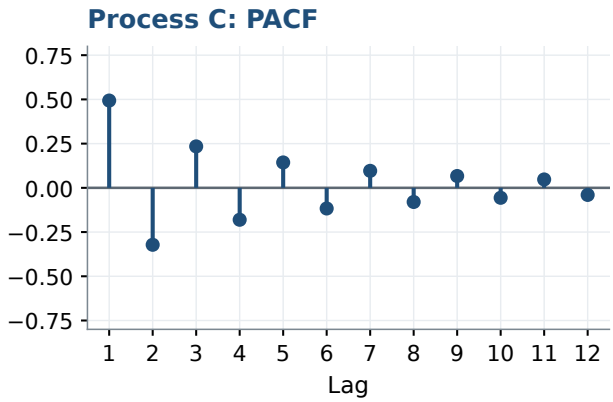


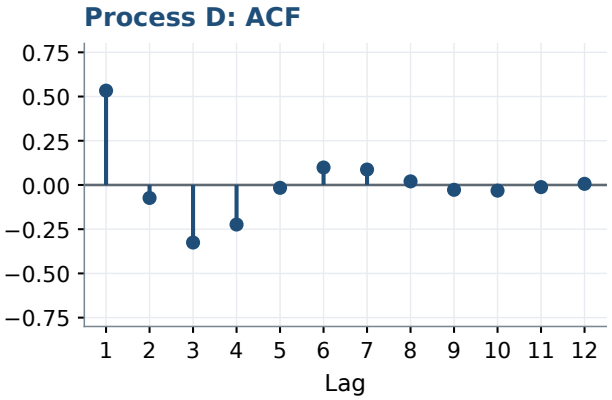


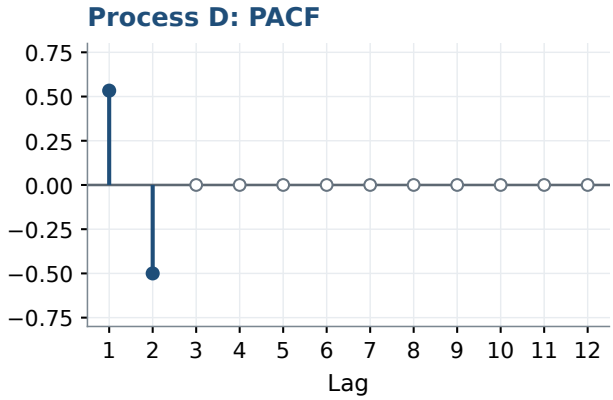


Exercise 2. Recognising AR and MA processes (continued)









Exercise 3. Unit roots: the EUR/USD exchange rate

Notebook: Unit_Root_Testing.ipynb

Analyse monthly EUR/USD levels and log changes for 2010–2025 (192 observations). Compare ADF tests with AIC and BIC lag selection, and assess the role of deterministic terms. Interpret the evidence against a unit root and its limits for conclusions about stationarity.

Relevant concepts

- Exchange-rate levels and log changes; quote direction; monthly aggregation.
- Unit roots; weak stationarity; stationarity around a deterministic trend.
- ADF null and alternative hypotheses; augmentation lags; deterministic terms.
- AIC versus BIC for lag selection; ADF test statistics, p-values and nonstandard critical values.
- Rejection versus failure to reject; finite-sample power; structural breaks and changing volatility.

Solutions

Exercise 1. Stationarity of ARMA models

Write the model as $\phi(L)y_t = \theta(L)\varepsilon_t$, where $Ly_t = y_{t-1}$. For the models here, which have no common AR and MA factors, a **causal weakly stationary solution** requires every root of $\phi(z) = 0$ to lie strictly outside the unit circle: $|z| > 1$. This gives a convergent representation in current and past innovations, consistent with the innovation assumption in the question.

(a) Here $\phi(z) = 1 - 0.5z$, whose root is $z = 2$. The root lies outside the unit circle, so a causal weakly stationary solution exists:

$$y_t = \sum_{j=0}^{\infty} (0.5)^j \varepsilon_{t-j}.$$

(b) Here $\phi(z) = 1 + 0.9z$, whose root is $z = -1/0.9 = -10/9$. Its modulus exceeds one, so a causal weakly stationary solution exists:

$$y_t = \sum_{j=0}^{\infty} (-0.9)^j \varepsilon_{t-j}.$$

Negative autoregressive dependence does not itself imply nonstationarity.

(c) The autoregressive polynomial factors as

$$\phi(z) = 1 + 1.9z + 0.88z^2 = (1 + 1.1z)(1 + 0.8z).$$

Its roots are $z_1 = -10/11$ and $z_2 = -5/4$. Since $|z_1| < 1$, there is **no causal weakly stationary solution** consistent with the stated innovation assumption. The MA polynomial is $\theta(z) = 1 + 0.5z$, with root -2. It shares no factor with $\phi(z)$, so the MA term cannot cancel the problematic AR root. Its root is outside the unit circle, so the MA component satisfies the invertibility condition; this does not repair the failure of causal stationarity.

The innovation assumption matters: roots inside the unit circle can sometimes be associated with stationary representations involving future shocks. Those noncausal representations are excluded here.

Exercise 2. Recognising AR and MA processes

- **Process A: MA(3).** The ACF is nonzero through lag 3 and exactly zero thereafter. The PACF tails off.
- **Process B: AR(3).** The PACF is nonzero through lag 3 and exactly zero thereafter. The ACF tails off.
- **Process C: MA(1).** Only lag 1 of the ACF is nonzero. The PACF tails off.
- **Process D: AR(2).** The PACF cuts off after lag 2, while the ACF tails off with a damped oscillation.

For an MA(q), the theoretical ACF cuts off after lag q; for an AR(p), the theoretical PACF cuts off after lag p. A tail may decay smoothly or oscillate. Thus, the ACF alone does not generally determine the AR order, but the ACF and PACF together identify it in these pure-model examples.

In a finite sample, estimated correlations fluctuate around their population values and do not show exact cutoffs. A few nonzero estimates beyond a proposed order may reflect sampling variation. Use the plots to suggest candidate models, then compare information criteria and check residual diagnostics; visual identification alone is not conclusive.